Assignment 2 Comp-4250

1)

E	S_1	S_2	S_3	S_4	h_1	h_2	h_3
0	0	1	0	1	1	2	2
1	0	1	0	0	3	5	1
2	1	0	0	1	5	2	0
3	0	0	1	0	1	5	5
4	0	0	1	1	3	2	4
5	1	0	0	0	5	5	3

$h_1(x) = 2x+1 \bmod 6$;
$h_2(x) = 3x+2 \bmod 6$;
$h_3(x) = 5x+2 \bmod 6$;

a) Signature matrix

	S_1	S_2	S_3	S_4
h_1	5	1	1	1
h_2	2	2	2	2
h_3	0	1	4	0

b) Only $h_3(x) = 5x+2 \bmod 6$ is a true permutation.

2)

a) $\max(x,y)$ = largest of x and y

This violates the second axiom $d(x,y) = 0$ if and only if $x = y$
This is only true for $\max(x,y)$ when x and y are 0.

If $x = 2$ and $y = 2$
then $m(2,2) = 2 \neq 0$

∴ max is not a distance measure.

b) $diff(x,y) = |x-y|$

i) $diff(x,y)$ is non-negative because $|x-y|$ will always be greater or equal to 0. As the definition of $|x|$.

$$|x| = \begin{cases} x, \text{if } x \geq 0 \\ -x, \text{if } x < 0 \end{cases}$$

ii) $diff(x,y) = 0 \iff x = y$ Proof

Let n be $n \in N$, $n \geq 0$

$diff(n,n) = |n-n| = |0|$
As per the definition of $|x|$ above, $|0| = 0$.
Since $n-n=0$ for any number n, $diff(x,y)$ Satisfies the second axiom.

iii) Prove Diff(x,y) is Symmetric

Case 1:
$x > y$; using definition of $|x|$

$diff(x,y) = |x-y| = x-y$

~~diff~~ $diff(y,x) = |y-x| = -(y-x)$
$= -y+x = x-y$

Both ways end with $x-y$

Case 2: $y > x$

$diff(x,y) = |x-y| = -(x-y) = -x+y$
$= y-x$

$diff(y,x) = |y-x| = y-x$

Both ways end with $y-x$

Case 3: $x = y$

$diff(x,y) = |x-y| = |0| = 0$

$diff(y,x) = |y-x| = |0| = 0$

Both end with 0

$\therefore$ diff(x,y) is symmetric

iv) Prove $diff(x,y) \leq diff(x,z) + diff(z,y)$

$|x-y| \leq |x-z| + |z-y|$
$\Rightarrow 0 \leq |x-z| + |z-y| - |x-y|$

Both absolute values and subtraction follow the triangle inequality
∴ so does $diff(x,y)$

∴ $diff(x,y)$ is a distance measure.

c) $Sum(x,y) = x+y$

This violates
$d(x,y)=0$ if and only if $x=y$

Ex; $Sum(2,2) = 2+2 = 4 \neq 0$

∴ $Sum(x,y)$ is not a distance measure.

3) L_1:

$|5-8|+|6-2|+|7-4|$
$=3+4+3=10$

∴ L_1 distance of $(5,6,7)$ and $(8,2,4)$ is 10.

L_2:

$(|5-8|^2+|6-2|^2+|7-4|^2)^{\frac{1}{2}}$
$=(3^2+4^2+3^2)^{1/2}$
$=(9+16+9)^{1/2}$
$=\sqrt{34}$
$\doteq 5.831$

∴ L_2 distance of $(5,6,7)$ and $(8,2,4)$ is $\sqrt{34}$

L_∞:

$\max(|5-8|, |6-2|, |7-4|)$
$=\max(3, 4, 3)$
$=4$

∴ L_∞ distance of $(5,6,7)$ and $(8,2,4)$ is 4.

4)

a) abcdef and bdaefc

abcdef $\xrightarrow{1}$ bcdef $\xrightarrow{2}$ bdef

$\xrightarrow{3}$ bdaef $\xrightarrow{4}$ bdaefc

∴ edit distance is 4.

b) abccdabc and acbdcab

abccdabc $\xrightarrow{1}$ accdabc

$\xrightarrow{2}$ acdabc $\xrightarrow{3}$ acbdabc

$\xrightarrow{4}$ acbdcabc $\xrightarrow{5}$ acbdcab

∴ the edit distance is 5.

c) abcdef and baedfc

abcdef $\xrightarrow{1}$ bcdef $\xrightarrow{2}$ bdef

$\xrightarrow{3}$ bdf $\xrightarrow{4}$ badf $\xrightarrow{5}$ baedf

$\xrightarrow{6}$ baedfc

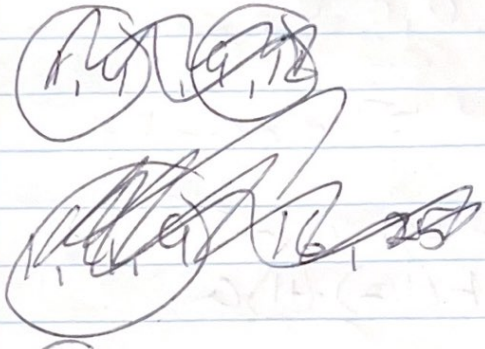
∴ the edit distance is 6.

Using L_1-norm distance

Initial data: Centroid = average

5) 1, 4, 9, 16, 25, 36, 49, 64, 81

~~First clusters:~~



Represent clusters by circles

2: (1, 4), 9, 16, 25, 36, 49, 64, 81

2: (1, 4, 9), 16, 25, 36, 49, 64, 81

is

3: (1, 4, 9), (16, 25), 36, 49, 64, 81

4: (1, 4, 9), (16, 25), (36, 49), 64, 81

5: (1, 4, 9, 16, 25), (36, 49), 64, 81

6: (1, 4, 9, 16, 25), (36, 49), (64, 81)

7: (1, 4, 9, 16, 25), (36, 49, 64, 81)

8: (1, 4, 9, 16, 25, 36, 49, 64, 81)

#4)

7)

a) $w = [0,0,0,0,0]$

$w \cdot a = 0 \Rightarrow$ move w

$w^{(t+1)} = w^t + \eta \cdot y^t \cdot x^t$

$w = w + (1/2)\cdot(+1)a$

$w = [0,0,0,0,0] + [\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}]$

$= [\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}]$

$w \cdot b = [\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}] \cdot [0,0,1,1,1]$

$= 1 \Rightarrow$ adjust w as $y = -1$

$w = w + (1/2)(-1)b$

$= [\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}] - [0,0,\frac{1}{2},\frac{1}{2},\frac{1}{2}]$

$= [\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, 0, 0]$

$w \cdot c = [\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, 0, 0] \cdot [0,1,1,0,0]$

$= 0 \Rightarrow$ adjust w as $y = +1$

$w = w + (1/2)(+1)c$

$= [\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, 0, 0] + [0, \frac{1}{2}, \frac{1}{2}, 0, 0]$

$= [\frac{1}{2}, 1, 0, 0, 0]$

$w \cdot d = [\frac{1}{2}, 1, 0, 0, 0] \cdot [1, 0, 0, 2, 0]$

$= \frac{1}{2} \Rightarrow$ adjust as $y = -1$

$w = w + (1/2)(-1)d$

$= [\frac{1}{2}, 1, 0, 0, 0] - [\frac{1}{2}, 0, 0, \frac{1}{2}, 0]$

$= [0, 1, 0, -\frac{1}{2}, 0]$

$w \cdot e = [0, 1, 0, -\frac{1}{2}, 0] \cdot [1, 0, 1, 0, 1]$

$= 0 \Rightarrow$ adjust w as $y = +1$

$w = w + (1/2)(+1)e$

$= [0, 1, 0, -\frac{1}{2}, 0] + [\frac{1}{2}, 0, \frac{1}{2}, 0, \frac{1}{2}]$

$= [\frac{1}{2}, 1, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}]$

$w \cdot f = [\frac{1}{2}, 1, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}] \cdot [1, 0, 1, 1, 0]$

$= \frac{1}{2} \Rightarrow$ adjust w as $y = -1$

$w = w + (1/2)(-1)f$

$= [\frac{1}{2}, 1, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}] - [\frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, 0]$

$= [0, 1, 0, -1, \frac{1}{2}]$

$\therefore w = [0, 1, 0, -1, \frac{1}{2}]$

b) $w = [1, 1, 1, 1, 1]$

$\theta = 5$

$w \cdot a = 4 \leq \theta$ and $x = +1$

$\therefore w = [2, 2, 1, 2, 2]$

$w \cdot b = 5 \geq \theta$ and $y = -1$

$\therefore w = [2, 2, \frac{1}{2}, 1, 1]$

$w \cdot e = 2.5 \leq \theta$ and $y = +1$

$\therefore w = [2, 4, 1, 1, 1]$

$w \cdot d = 3 \leq \theta$ and $y = -1$

$\therefore$ w remains the same

$w \cdot e = 4 \leq \theta$ and $y = +1$

$\therefore w = [4, 4, 3, 2, 2]$

$w \cdot f = 7 \geq \theta$ and $y = -1$

$\therefore w = [2, 4, 1, \frac{1}{2}, 2]$

C still doesn't agree

$w \cdot c = 5 \not\geq \theta$ and $y = +1$

$\therefore w = [2, 8, 2, \frac{1}{2}, 2]$

now everything converges

$\therefore w = [2, 8, 2, \frac{1}{2}, 2]$

c) $w = [0, 0, 0, 0, 0, 0]$

$w \cdot a = 0$ and $y = +1$

$w = w + \frac{1}{2}a$

$= [\frac{1}{2}, \frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}]$

$w \cdot b = 1.5$ and $y = -1$

$w = w - \frac{1}{2}b$

$= [\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, 0, 0, 0]$

$w \cdot c = 0$ and $y = +1$

$w = w + \frac{1}{2}c$

$= [\frac{1}{2}, 1, 0, 0, 0, -\frac{1}{2}]$

$w \cdot d = 1$ and $y = -1$

$w = w - \frac{1}{2} d$

$= [0, 1, 0, -\frac{1}{2}, 0, 0]$

$w \cdot e = 0$ and $y = +1$

$w = w + \frac{1}{2} e$

$= [\frac{1}{2}, 1, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}]$

$w \cdot f = 1$ and $y = -1$

$w = w - \frac{1}{2} f$

$= [0, 1, 0, -1, \frac{1}{2}, 0]$

$\therefore w = [0, 1, 0, -1, \frac{1}{2}, 0]$

d) $w = [1, 1, 1, 1, 1, 1]$

$w \cdot a = 3 > 0$ and $y = +1$

Do nothing

$w \cdot b = 2 > 0$ and $y = -1$

So, $w = [1, 1, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}, 2]$

w.c = -0.5 < θ and y = +1

So, w = [1, 2, 1, ½, ½, 1]

w.d = 0.5 < θ and y = -1
So do nothing

w.e = 1.5 > θ and y = +1
So do nothing

w.f = 1.5 > θ and y = -1

So, w = [½, 2, ½, ¼, ½, 2]

w.a = 1.25 < θ and y = +1

So w = [1, 4, ½, ½, 1, 1]

w.b = 1 = θ and x = -1

So, w = [1, 4, ¼, ¼, ½, 2]

w.c = 2.25 > θ and y = +1
So do nothing, same w/ w.d

w.e = -0.25 < θ and y = +1

So, w = [2, 4, ½, ¼, 1, 1]

w.f=1.75 > 0 and x=-1

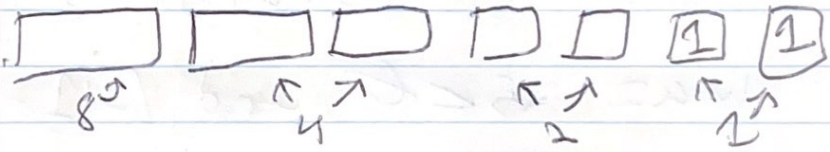
So, w2 $[1, 4, 1, \frac{1}{2}, 1, \frac{1}{3}]$

Either it doesn't converge
Or I started doing it wrong



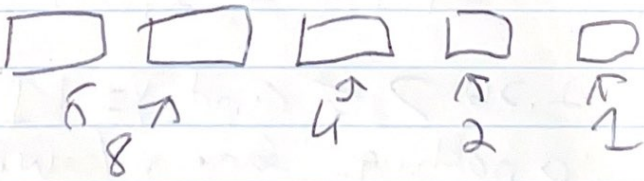
skip to 11

11) First 1 enters



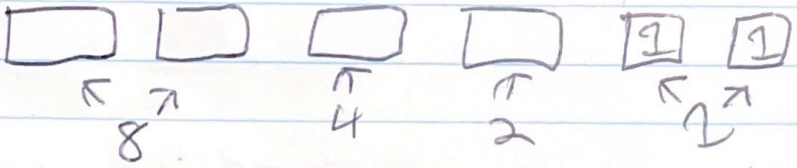
~~AA~~ A new bucket of size 1 is created

Second 1 enters



Second 1 causes a chain reaction of merges so that 2 8-buckets, 1 4-bucket, 1 2-bucket and 1 1-bucket appears

third 1 enters:



The third 1 does not cause any merges as 2 same type buckets are allowed.